

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

R2 is only forming an OSPF neighbor relationship with R1 because only interface S0/0/0 is active for OSPF. The interface S0/0/1 is passive, so R2 does not exchange OSPF routing updates with R3.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? S0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65: Cost = 64 (Serial link) + 1 (Gigabit Ethernet)

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because modern networks use link speeds faster than 100 Mbps.

The default OSPF reference bandwidth is 100 Mbps, which would cause fast links such as Gigabit Ethernet and 10-Gigabit Ethernet to have the same cost. Changing the reference bandwidth allows OSPF to calculate more accurate routing costs. _____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

__ The OSPF cost is calculated using the formula: $\text{Cost} = \text{Reference Bandwidth} / \text{Interface Bandwidth}$ __

__ The cost to the 192.168.3.0/24 network is calculated by adding the cost of R1 S0/0/1 (781) and R3 G0/0 (1), which equals **782**.__

__ The cost to the 192.168.23.0/30 network is calculated by adding the cost of R1 S0/0/1 (781) and R3 S0/0/1 (64), resulting in a total cost of **845**.__

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

__ 1562. The serial interface bandwidth was changed to 128 kbps, increasing the OSPF interface cost.

Since the route from R1 to the 192.168.23.0/30 network passes through two serial links, the accumulated cost is: $781 + 781 = 1562$ __

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

___ OSPF always selects the route with the lowest accumulated cost. After manually increasing the OSPF cost on interface S0/0/1 of R1 using the ip ospf cost command, the direct route from R1 to R3 became more expensive. The accumulated cost of the route through R2 is: R1 S0/0/0 (781) + R2 S0/0/1 (781) + R3 G0/0 (1) = 1563. The accumulated cost of the direct route from R1 to R3 is: R1 S0/0/1 (1565) + R3 G0/0 (1) = 1566. Since 1563 is lower than 1566, OSPF selects the route through R2, so the traffic to the 192.168.3.0/24 network is now routed via R2. ___

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

___ The router ID uniquely identifies each router in the OSPF network. Controlling the router ID ensures stable OSPF operation and predictable neighbor relationships. If router IDs change unexpectedly, OSPF adjacencies may reset and routing recalculations may occur. ___

2. Why is the DR/BDR election process not a concern in this lab?

___ The DR/BDR election occurs only on multi-access networks such as Ethernet. In this lab, the routers are connected using point-to-point serial links, which do not require a DR/BDR election. ___

3. Why would you want to set an OSPF interface to passive?

___ A passive interface prevents the router from sending OSPF hello packets on that interface. This reduces unnecessary routing traffic and improves security on networks where no OSPF neighbors exist, such as LANs connected to end devices. ___

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

___ OSPF uses a hierarchical design based on areas. The use of multiple areas, connected through the backbone area (Area 0), allows large networks to be divided into smaller segments, improving scalability and reducing routing overhead. _

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

___ network 10.0.0.0 0.255.255.255 area 0 ___

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

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The router will select the EIGRP route. Routing protocols are selected based on Administrative Distance, not metric values. EIGRP has the lowest administrative distance, its route will be chosen. _____